MINERVA CryoCell P&ID — Reproduction & Standardisation Report

Project: MINERVA CryoCell (SCK CEN / 84836013)
Deliverable: Reproduced & improved Piping & Instrumentation Diagrams
Standards applied: ISO 10628 (flow diagrams) · ANSI/ISA-5.1 (instrumentation symbols & identification)
Sheet format: ISO A3 landscape — 420 mm × 297 mm
Version: v2.0 (rebuilt)

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1. Scope

Two source P&ID sheets were reproduced and improved:

Key	Source file	Title
QCELL-LB	`PFD-PID MINERVA QCELL-LB.svg`	QCELL / LB cryogenic flow scheme (full)
RFCELL	`PFD-PID MINERVA RFCELL seen by ACR.svg`	RFCELL DI-water / coupler module

The rebuild is **data-driven**: it re-uses the validated segmentation produced in the previous project stage (`segmentation/data/*.json`) for all instrument, equipment, safety and vacuum elements, and re-extracts the original process-line geometry with fully resolved transforms so the pipe network is preserved exactly while being recoloured and re-weighted to standard.

Output files (`output/`)

File	Description
`PID_QCELL-LB_improved.svg` / `.pdf`	Improved QCELL-LB sheet (A3)
`PID_RFCELL_improved.svg` / `.pdf`	Improved RFCELL sheet (A3)
`TECHNICAL_DOCUMENTATION.md`	This report
`LAYER_STRUCTURE.md`	Layer hierarchy reference
`previews/*.png`	Raster previews

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2. How the diagrams were rebuilt

The generator lives in `generator/`:

Module	Role
`svg_extract.py`	Walks the source SVG, accumulates the full CTM for every node, captures every graphic primitive with resolved transform, stroke/fill colour, width, dash and owning layer, and **bins** it (process line by class, structure, instrument-bubble candidate, dashed boundary, coloured fill/node).
`symbols.py`	ISA-5.1 / ISO-10628 symbol primitives: instrument bubbles, valves (manual / control / solenoid / relief), vessels, cavity/coupler bodies, heat exchangers, terminal points, heat-load callouts, junction nodes.
`build_pid.py`	Composes the A3 sheet: frame + title block, equipment, colour-segmented process lines, vacuum barriers, fresh ISA instrument bubbles, ISA tags and the full legend. Emits SVG and feeds the PDF conversion.

Re-run with:

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cd generator && python3 build_pid.py
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3. Improvements made

3.1 Sheet, border & title block

- Re-laid out on a **true ISO A3 landscape** canvas (420 × 297 mm, `viewBox 0 0 1587.273 1122.430`, ratio 1.41414).
- Added a **double-line drawing border** and a structured **title block** (project, title, drawing number, standard, revision).
- The original drawing content is mapped uniformly into a reserved content region so the border, the right-hand legend column and the bottom class-legend band never overlap the schematic.

3.2 Layer hierarchy (ISO logical structure)

The drawing is organised into the requested 7-level Inkscape layer hierarchy with process lines split into colour/class sub-layers — see `LAYER_STRUCTURE.md`.

3.3 ISA-5.1 instrumentation

- Every field instrument is redrawn as a **standard circular bubble** with a **two-line tag** (measured-variable + function letters over loop number), e.g. `TT / 514`.
- **Tag grammar** follows ISA-5.1: first letter = measured variable, succeeding letters = function:
- `T` Temperature, `P` Pressure, `L` Level, `F` Flow, `E` Electrical, `A` Analysis, `S` Speed/Safety, `R` Radiation …
- `T` Transmitter, `I` Indicator, `S` Switch, `V` Valve, `E` Element …
- Examples present: `TT` temperature transmitter, `PT` pressure transmitter, `LT` level transmitter, `FT` flow transmitter, `LS` level switch, `EH` electric heater, `PZ/FZ` safety variants, `AP` analysis/antenna probe.
- **Bubble fill encodes the instrument family** (per the source legend): white = LB cryo instrument, salmon = RFCELL instrument, light-blue = LBI-specific instrument.
- **Protection / safety instruments** (`SV`, `RV`, `PL`, `PZ`, `FZ` …) use a **dashed bubble outline** — the ISA convention for interlock/safety functions.
- **Valves** are drawn with correct bodies and actuators: hand valve (hand-wheel), control valve (diaphragm), solenoid valve (`S` box), relief valve (angle/spring) — replacing generic bubbles for `HV/CV/SV/RV`.

3.4 Colour segmentation of process lines

The original used inconsistent / very light screen colours. Lines are now recoloured to a **consistent, print-legible palette**, one Inkscape sub-layer per class, with standard weights and line styles:

Class	Service	Colour	Line style
A	4.5 K / 3 bar — LHe supply	blue `#0033cc`	solid, heavy
B	3.5 K / 27 mbar — 2 K LP return	teal `#00a6bd`	solid, heavy
D	40 K / 14 bar — thermal shield	red `#e00000`	solid, heavy
E	60 K / 13 bar — return / He-guard	olive `#8a8a00`	solid, heavy
Water	DI cooling water (coupler / FREIA)	green `#00a000`	solid, medium
QINFRA	infrastructure scope division	dark green `#006400`	dashed
Inst. air	pneumatic 6 (5–7) bar(g)	magenta `#c000c0`	dash-dot

Bulky solid colour blocks in the source were converted to **translucent service zones** (22 % fill) so they read as highlighting rather than obscuring the schematic; pipe junctions are drawn as small solid nodes in the class colour.

3.5 Vacuum barriers & boundaries

Dashed structural boundaries (vacuum vessel / scope divisions) are collected into a dedicated **L4 — Vacuum Barriers & Boundaries** layer with a standard dashed style, plus the explicit “vacuum barrier” annotation.

3.6 Legend enhancement

A comprehensive two-part legend was added:
- **Right column:** instrument-bubble key (families + safety), tag-grammar note, valve & equipment symbols, heat-load callout, terminal/scope point, vacuum boundary.
- **Bottom band:** process-line class legend mapping each colour to its temperature/pressure service and description.

3.7 Quality

- Standardised line weights and rounded joins/caps.
- Minimum on-sheet text sizes kept legible (≈ 6 pt and up).
- Suppressed the messy/overlapping original bubble circles and redrew clean ones.
- Markers (arrowheads) and gradient/def references from the source are carried over so flow-direction arrows are preserved.

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4. Element inventory (from segmentation)

Metric	QCELL-LB	RFCELL
Instrument tags	230 (221 unique)	67 (45 unique)
Equipment items	42	2
Safety devices	13	6
Vacuum barriers	5	0
Temperature points	77	23
Pressure points	28	11
Process-line classes	A,B,D,E,Water,QINFRA,Air	A,B,D,E,Water,QINFRA,Air

Instrument prefixes encountered (ISA-5.1):
`TT, TE, PT, PI, PZ, LT, LS, LI, FT, FI, FZ, EH, SM, RS, AP, AA, AD, ED, CV, HV, SV, RV, PL, KW, AK, MV, HL, HX, CF`.

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5. Limitations & notes

- The source sheets are intentionally **very dense**; the rebuild preserves the real topology and instrument positions rather than re-routing pipes, so localised crowding inherent to the originals remains.
- Colour-to-class mapping is taken from the drawing legend; a few low-confidence source colours (grey shading, near-black) are treated as structure.
- Geometry fidelity is exact (transforms are baked); symbology and styling are the standardised improvement layer.